

**CS7NS6 Distributed Systems
2025-2026
Exercise 2**

Implementing a Globally-accessible Distributed Service

Final Report Outline

The final report should include chapters addressing the following issues:

1. **Requirements:** describe the functionality provided and outline the functional and non-functional requirements addressed in its design.
2. **Specification:** provide a detailed specification of the functionality provided by your system including the services of which it is composed and their API(s) as well as a description of how it behaves in the presence of various types of faults. Be sure that this description is consistent with what you will demonstrate, i.e., this description should note any of the requirements that the system does not (fully) address.
3. **Architecture and design:** provide a description of your system's technical and software architectures. Describe the runtime services that are provided and how they are distributed/connected.
Each member of the team should provide a description of the design of the single most significant service for which they were responsible.
4. **Implementation:** provide a description of the operation of the system including behavioural diagrams for important algorithms. Describe how the system handles important failure modes. Describe important algorithms such as those for member failure detection, achieving consensus on group membership, and tolerating partitions.
5. **Testing:** outline your test plan and the results of your testing.
6. **Allocation of work:** describe how the work was distributed among team members.
7. **Summary:** summarise your achievements.

The report should be no longer than 20 pages (and may be significantly shorter). Good use of explanatory diagrams is recommended. The report should be signed by all members of the group.